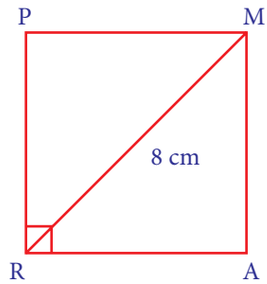
5.16.2 Construction of a square when its diagonal is given

Example 5.41

Construct a square RAMP of a diagonal 8 cm. Also find its area.

Solution:

Given: diagonal = 8 cm



Rough diagram

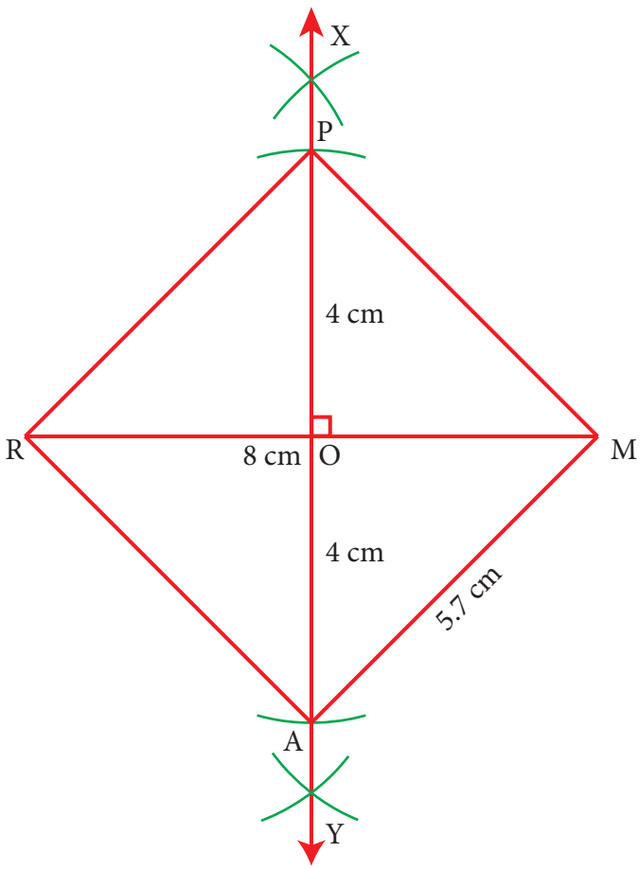


Fig. 5.60

Steps:

1. Draw a line segment RM = 8 cm.
2. Draw the perpendicular bisector XY to RM. Let it bisect RM at O.
3. With O as centre, draw arcs of radius 4 cm on either side of O which cut OX at P and OY at A.
4. Join RA, AM, MP and PR.
5. RAMP is the required square.

Calculation of Area:

Area of square RAMP $= a^2$ *sq.units*

$= 5.7 \times 5.7 = 32.49$ *sq.cm*

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